

SBC 201 Chapter 19 Concrete — Group R-2 High-Rise Dimensional Requirements Matrix (2024)

1. Document metadata and use limitation

- **Project basis:** Riyadh, Saudi Arabia; Group R-2 residential high-rise; an occupied floor is stated to be more than 23 m above the relevant reference level.
- **Deliverable tier:** Project-use matrices in Sections 1–12 (design-check rows, not pasted inventory), plus a coverage summary and unresolved-source register. The full row inventory is not published.
- **Code/source basis:** SBC 201 (2024), Chapter 19, source file Reference\SBC 201 2024\source_reference\Chapter_19 – CONCRETE.txt.
- **Extraction audit:** Skill-finetune re-run. Project-use rows follow the chapter-extract row contract (noun-phrase checks, bold SI values, building-language triggers, named exceptions, check-specific actions). Internal inventory: **33** independently checkable numeric records (**33** Verified, **0** Verify source). Non-numeric OCR hold points are listed in the register and are not design-release values. No pre-skill Chapter 19 baseline is published beside this file.
- **Model:** Cursor Grok 4.6.
- **Prepared:** 2026-08-27.
- **Status:** Source-only architectural advisory matrix for design coordination. It is not a stamped compliance statement, structural design, SCD NOC, SBPS approval, legal opinion, or permission to construct.
- **Outbound-source rule:** No value in this matrix has been imported from SBC 304, SBC 301, SBC 302, Chapter 17, Section 2206, Chapter 7 / Table 721.1(2), ACI 117, ACI ITG-7, ACI 506R, PCI MNL 128, ASTM E2634, commentary examples, or the existing chapter summary. Where Chapter 19 sends the user elsewhere, this matrix records the dependency without supplying the outbound value.

Scope and assumptions

1. Group R-2 and high-rise status are project statements, not independently verified classifications.
2. The exact Riyadh AHJ/permit pathway, project stage, fire-strategy status and SCD NOC status are unconfirmed; therefore this matrix does not conclude compliance.
3. Seismic Design Category is unconfirmed. Item 11 of 1901.5 (slab-on-grade as a structural diaphragm) applies only if the structure is assigned to SDC **D, E or F**.
4. Automatic sprinkler protection is not selected in this chapter extract. Chapter 19 does not branch on NFPA 13 versus 13R.
5. Building height, storey count, grade plane, basement/podium construction, mixed use and whether any slab on grade transmits vertical loads or lateral forces are unconfirmed.
6. Structural concrete design, durability classes, anchoring, materials and most member sizes live in **SBC 304**, not in this chapter. Section 1905 states that appropriate modifications have been reflected in **SBC 304-24**; this extract does not reprint those modifications.
7. 1901.7.1 says cast-in-place tolerances shall be “in accordance with this section” but the attached extract does not publish millimetre tolerances or name ACI 117 in the **code** paragraph (ACI 117 appears only in commentary). No commentary millimetres are adopted.
8. There are no appended tables in this chapter extract.

2. Legends

Applicability

Status	Meaning
Direct	Expected to govern the stated R-2 tower basis, subject to confirmed geometry and design data.
Conditional	Governs only when the stated feature, load, sprinkler branch or exception exists.
Not typical	Unrelated occupancy-only rule; omitted from this deliverable unless the gap register already opened that use.
External verification	Chapter 19 points to another section/code/standard, or the project/AHJ basis must be confirmed before use.

Source confidence

Status	Meaning
Verified	Requirement and any stated numeric value were checked against unambiguous mandatory Chapter 19 source text.
Verify source	OCR, flattened table, page-split, or footnote attachment is unresolved. Not a design-release value.

3. Project decision and gap register

Decision / gap	Current project basis	Why it controls Chapter 19 application	Required project action
Structural engineer / SBC 304	Unconfirmed companion design	1901.2 charges structural concrete to this chapter and SBC 304 as amended in 1905; this extract has no member sizes, covers, mix classes or anchor capacities	Issue an SBC 304 structural package; do not size concrete from Chapter 19 alone
Seismic Design Category	Unconfirmed	1901.5 Item 11 requires a statement whether slab on grade is a structural diaphragm for SDC D, E or F ; 1902.1.1 design displacement is outbound to SBC 301 12.8.6	Lock SDC on the code datum sheet; show diaphragm/non-diaphragm SOG on structural drawings
Slab on grade role	Unconfirmed: architectural wearing slab vs structural diaphragm / collector	Non-structural SOG is limited here to 1904 and 1907; a load-path SOG must comply with the full chapter and SBC 304	Structural engineer to classify every ground-supported slab
1906.1 light-frame footings	Not applicable to this R-2 high-rise	Plain-concrete footing 150 mm / 100 mm projection is only for Group R-3 and other occupancies less than two stories of light-frame construction	Do not apply 1906.1 to tower or typical podium foundations
Shotcrete use	Unconfirmed	Section 1908 numeric locks apply only if shotcrete is specified	Confirm whether shotcrete is used (repair, basement, landscape); if none, park 1908 as unused
Vapor-retarder AHJ exception	Unconfirmed	1907.1 Exception 2 waives the retarder where approved based on local site conditions	Do not omit the retarder under enclosed floors without AHJ approval
Special inspections	Outbound to Chapter 17	1901.6 and 1903.2 send concrete SI/tests to Chapter 17; this extract has no SI frequency table	Coordinate Chapter 17 SI schedule with the concrete specification
CIP tolerance standard	1901.7.1 charging incomplete in the extract	Code does not name ACI 117; commentary does. No millimetre tolerances are published here	Verify the published 1901.7.1 text before specifying CIP tolerances

Decision / gap	Current project basis	Why it controls Chapter 19 application	Required project action
Mixed-use podium	Unconfirmed	Does not change 1907 SOG thickness; may add GFRC cladding (1903.3) or ICF walls (1903.4)	Freeze cladding and forming systems; import PCI MNL 128 / ASTM E2634 only if used
NOC / stamped structural	Unconfirmed	Concrete design, SCD acceptance and construction tolerances cannot be concluded from this chapter extract	Engage the structural engineer of record before design freeze

4. Scope and SBC 304 companion

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1901.1	Chapter concrete scope	This chapter governs materials, quality control, design and construction of concrete used in structures	Concrete in the structure	None stated	Direct	Treat Chapter 19 as the charging chapter; send member design to the SBC 304 package	Verified
1901.2	Structural concrete companion	Structural concrete shall be designed and constructed in accordance with this chapter and SBC 304 as amended in 1905	Structural concrete	Slabs on grade that do not transmit vertical loads or lateral forces are governed here only by 1904 and 1907	Direct	Keep a live SBC 304 structural specification; classify each SOG as structural or non-structural	Verified
1901.3	Anchoring to concrete	Anchoring to concrete shall be in accordance with SBC 304 as amended in 1905	Cast-in, post-installed expansion, undercut, screw and adhesive anchors	None stated	External verification	Schedule all concrete anchors to the SBC 304 anchorage specification; do not invent capacities here	Verified
1901.4	Composite steel-concrete design	Systems of structural steel acting compositely with reinforced concrete shall be designed in accordance with Section 2206	Composite steel-concrete systems	None stated	External verification	If composite slabs or beams are used, design them under Section 2206, not from this chapter	Verified
1905.1	SBC 304-24 modifications	Appropriate modifications has been reflected in SBC 304-24	Structural concrete using SBC 304	None stated	External verification	Use the SBC 304-24 text as modified; do not reconstruct 1905 amendments from this extract	Verified

5. Construction documents and inspections

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1901.5	Concrete construction-document contents	Construction documents for structural concrete shall include the 1901.5 strength, grade, geometry, prestress, splice, joint and outbound SBC 301 / SBC 304 Chapter 26 information	Structural concrete construction documents	Item 11 is a separate SDC D, E or F statement	Direct	Put the 1901.5 checklist on the structural issue set	Verified
1901.5 Item 11	SDC slab-diaphragm statement	For structures assigned to Seismic Design Category D, E or F , the documents shall include a statement if slab on grade is designed as a structural diaphragm	SDC D, E or F	Not required by this item outside SDC D, E or F	Conditional	Once SDC is locked, state on the foundation drawings whether each SOG is a diaphragm	Verified
1901.6	Concrete special inspections	Special inspections and tests of concrete elements and concreting operations shall be as required by Chapter 17	Concrete elements and concreting operations	None stated in this chapter	External verification	Attach the Chapter 17 SI schedule to the concrete specification; do not invent frequencies here	Verified
1903.2	Materials special inspections	Where required, special inspections and tests shall be in accordance with Chapter 17	Concrete materials and testing where SI is required	None stated in this chapter	External verification	Align material testing with the same Chapter 17 programme as 1901.6	Verified

6. Tolerances

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1901.7	Default structural tolerances	Where not indicated in construction documents, structural tolerances for concrete structural elements shall be in accordance with 1901.7	Concrete structural elements without stated tolerances	None stated	Direct	State CIP and precast tolerances on the drawings so 1901.7 is not an unspoken default	Verified
1901.7.1	Cast-in-place concrete tolerances	Structural tolerances for cast-in-place concrete structural elements shall be in accordance with this section; the attached extract publishes no millimetre limits and does not name ACI 117 in the code paragraph	Cast-in-place concrete structural elements	Group R-3 detached one- or two-family dwellings; shotcrete	External verification	Verify the published 1901.7.1 text before specifying CIP tolerances; do not import ACI 117 millimetres from commentary	Verify source
1901.7.2	Precast concrete tolerances	Structural tolerances for precast concrete structural elements shall be in accordance with ACI ITG-7	Precast concrete structural elements	Group R-3 detached one- or two-family dwellings are not required to comply	External verification	If precast is used, specify ACI ITG-7; do not copy ITG-7 values into this matrix	Verified

7. Terminology and special structural walls

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1902.1.1	Design-displacement definition	Design displacement at each level shall be the total lateral deflection at that level calculated for the design earthquake using Section 12.8.6 of SBC 301	Seismic design displacement	None stated	External verification	Take storey displacements from the SBC 301 analysis; do not invent millimetres here	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1902.1.2	Special structural wall	Special structural walls of cast-in-place or precast concrete shall comply with SBC 304 Sections 18.2.4 through 18.2.8, 18.10 and 18.11 , as applicable, in addition to ordinary wall requirements; where SBC 301 says “special reinforced concrete shear wall,” it means “special structural wall”	Special structural walls	Ordinary wall requirements still apply as applicable	External verification	Match wall type labels on structural drawings to SBC 304 special-wall clauses; do not import those clause values here	Verified

8. Materials, GFRC and ICF

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1903.1	Concrete materials and tests	Materials used to produce concrete, concrete itself and testing thereof shall comply with the applicable standards listed in SBC 304	Concrete materials and testing	None stated	External verification	Specify cements, aggregates, water, reinforcement and admixtures from the SBC 304 list	Verified
1903.3	GFRC materials standard	Glass fiber-reinforced concrete (GFRC) and the materials used in such concrete shall be in accordance with PCI MNL 128	GFRC	None stated	Conditional	If GFRC cladding or panels are used, specify PCI MNL 128; do not take fire ratings from this chapter	Verified
1903.4	Flat-wall ICF material	Insulating concrete form material used for forming flat concrete walls shall conform to ASTM E 2634	Flat-wall ICF systems	None stated	Conditional	If ICF walls are proposed, require ASTM E 2634 forms; typical R-2 high-rise CIP/precast does not use ICF	Verified

9. Durability

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1904.1	Structural concrete durability	Structural concrete shall conform to the durability requirements of SBC 304	Structural concrete	None stated	External verification	Assign SBC 304 exposure classes on the mix schedule; do not invent cover or w/c limits here	Verified
1904.2	Nonstructural concrete durability	The registered design professional shall assign appropriate durability requirements for nonstructural concrete	Nonstructural concrete	None stated	Direct	State durability for toppings, non-structural SOG and similar work on the specification, even though SBC 304 does not assign them here	Verified

10. Slabs on grade

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1907.1	Ground-supported slab and vapor retarder	Concrete floor slabs supported directly on the ground shall be not less than 90 mm thick, with a 150 microns (0.15 mm) polyethylene vapor retarder, joints lapped not less than 150 mm , between base/subgrade and slab, or other approved equivalent methods or materials	Concrete floor slabs on the ground	Vapor retarder is not required for driveways, walks, patios and other flatwork that will not be enclosed later; or where approved based on local site conditions. Thickness has no stated exception	Direct	Show 90 mm SOG with 150-micron retarder and 150 mm laps on enclosed ground-floor, basement and parking details	Verified

11. Shotcrete

Apply this section only if shotcrete is specified. Except as modified here, shotcrete shall conform to this chapter for plain or reinforced concrete (1908.1). Do not adopt commentary millimetres (ACI 506R **50 mm** WWF, **750 mm / 75 mm** preconstruction panels, **300 mm** joint taper, **66°C** steam, **5°C** curing).

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1908.1–1908.2	Shotcrete proportions and strength	Shotcrete is mortar or concrete pneumatically projected at high velocity onto a surface; proportions shall allow suitable placement with the selected equipment and shall result in finished in-place hardened shotcrete meeting the strength requirements of this code	Shotcrete work	1908 modifies this chapter; otherwise plain/reinforced concrete rules apply	Conditional	If shotcrete is used, submit a mix that meets the specified strength under 1908.10 testing	Verified
1908.3–1908.4.3	Shotcrete aggregate and reinforcement	Coarse aggregate, if used, shall not exceed 19 mm . Maximum reinforcement size is Dia 16 bars unless preconstruction tests demonstrate adequate encasement of larger bars. Dia 16 or smaller: minimum clearance between parallel bars 65 mm . Larger bars where permitted: minimum clearance six diameters . Two curtains: nearer-nozzle curtain 12 bar diameters , remaining curtain six bar diameters . Lap splices shall be noncontact with 50 mm minimum clearance between bars	Shotcrete reinforcement placement	Building official may reduce required clearances where preconstruction tests show adequate encasement. Contact lap splices for support are permitted when approved from satisfactory preconstruction tests and the splice plane is perpendicular to the shotcrete surface	Conditional	Dimension bar size, parallel clearance and noncontact laps on the shotcrete placing drawing	Verified
1908.4.4	Shotcrete spirally tied columns	Shotcrete shall not be applied to spirally tied columns	Spirally tied columns	None stated	Conditional	Keep spiral columns as CIP or another permitted method; do not gun them	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1908.5	Shotcrete preconstruction tests	Where required by 1908.4, a test panel shall be shot, cured, cored or sawn, examined and tested before the project starts; the panel shall reproduce the thickest and most congested area, shot at the same angle with the same nozzle, mix and equipment unless substitutes are approved; reports to the building official as specified in Section 1704.5	Preconstruction tests required by 1908.4	Substitute equipment only if approved by the building official	Conditional	Do not import commentary 750 mm / 75 mm panel sizes; follow 1908.5 as written and 1704.5 for reporting	Verified
1908.6–1908.8	Shotcrete placement and joints	Rebound or accumulated loose aggregate shall be removed before placing initial or succeeding layers; rebound shall not be used as aggregate. Unfinished work shall not stand for more than 30 minutes unless edges are sloped to a thin edge. Square joints are permitted for structural elements that will be under compression and for construction joints shown on the approved construction documents. Sloping and square edges shall be cleaned and wetted before placing additional material. In-place shotcrete with sags, sloughs, segregation, honeycombing, sand pockets or other obvious defects shall be removed and replaced; shotcrete above sags and sloughs shall be removed and replaced while still plastic	Shotcrete placement	Square joints only where 1908.7 permits them	Conditional	Limit cold joints to 30 minutes or a sloped thin edge; do not reuse rebound; replace defective shotcrete. The extract’s “Before placing additional material are permitted” sentence is OCR-garbled — do not invent missing words	Verified
1908.9–1908.9.2	Shotcrete curing	During the curing periods specified, shotcrete shall be maintained above 4°C and in moist condition. Initial curing: continuously moist for 24 hours after shotcreting is complete, or sealed with an approved curing compound. Final curing shall continue for seven days after shotcreting, or for three days if high-early-strength cement is used, or until the specified strength is obtained, using the initial-curing process or an approved moisture-retaining cover	Shotcrete curing	Three-day final-curing branch only with high-early-strength cement, or stop when specified strength is obtained	Conditional	Put 4°C, 24 hours and seven days / three days on the shotcrete curing specification; do not substitute commentary 5°C or 66°C	Verified
1908.10–1908.10.3	Shotcrete strength tests	Approved-agency tests on representative specimens water-soaked at least 24 hours before testing. Maximum-size aggregate larger than 9.5 mm : not less than three 75 mm cores or 75 mm cubes; test panels minimum 450 mm by 450 mm . Aggregate 9.5 mm or smaller : not less than 50 mm cores or 50 mm cubes; test panels minimum 300 mm by 300 mm . Sampling at least once each shift , but not less than one for each 40 m³ . Average of three cores $\geq$ 0.85f’c with no single core $<$ 0.75f’c ; average of three cubes $\geq$ f’c with no individual cube $<$ 0.88f’c ; retest erratic locations	Shotcrete strength acceptance	Specimens from in-place work or test panels. Panels shot in the same position, during the work, by the nozzlemen doing the work, and cured as the work	Conditional	Write the 9.5 mm specimen/panel split, soak, frequency and 0.85f’c / 0.75f’c / 0.88f’c acceptance into the QA specification	Verified

12. Project-use controls

1. Use **Verified** rows for initial scoping after the row trigger and branch are confirmed.
2. Treat every **Verify source** row as a design hold point; no affected value is to be placed in issued-for-approval drawings without a published-source check.
3. Do not apply 1906.1 **150 mm / 100 mm** plain-concrete footings to this R-2 high-rise.
4. Do not import SBC 304 covers, mix classes, anchor capacities, ACI 117 millimetres, ACI ITG-7 values, PCI MNL 128 details, ASTM E2634 tests, Chapter 17 SI frequencies, or commentary examples (**750 mm** panels, **50 mm** WWF, **300 mm** tapers, **66°C, 5°C**).
5. Keep the published tokens **150 microns (0.15 mm), 4°C, Dia 16, 0.85f’c, 0.75f’c** and **0.88f’c** intact.
6. Record SDC, SOG structural role, shotcrete use and vapor-retarder AHJ decisions in the project Golden Thread; this matrix is not evidence of SCD NOC or stamped compliance.

13. Coverage summary

Internal inventory of the attached Chapter 19 extract (numbered code, exceptions, tables, footnotes; commentary excluded). Row-level records are not published.

- **Inventory scope:** numbered code, exceptions, tables, footnotes (commentary excluded)
- **Total independently checkable numeric records:** 33
- **Verified:** 33
- **Verify source:** 0
- **Appended tables:** none

Counts by top-level section

Top-level section	Records
1901	0
1902	0
1903	0
1904	0
1905	0
1906	3
1907	3
1908	27

Coverage cross-check against SBC 201 Chapter 19 Concrete (2024)_CS.md was topics-only: SBC 304 companion; SOG 1904/1907 versus structural SOG; 1906 light-frame footings not applied to R-2 high-rise; 1907 **90 mm / 150 microns / 150 mm** lap; GFRC fire ratings are not this chapter. No CS.md value was copied into a matrix cell.

14. Unresolved-source register

Hold points for project-use **Verify source** rows and OCR clauses that must not be repaired from memory. No value in this register is a design-release figure. Numeric inventory **Verify source** count remains **0**.

Affected table / clause	Why unverified	Control note
1901.7.1	Code says CIP tolerances shall be "in accordance with this section" but the extract publishes no millimetre values and does not name ACI 117 in the code paragraph; ACI 117 appears only in commentary	Verify the published 1901.7.1 text. Do not import ACI 117 millimetres from commentary or memory
1908.7	After the square-joint permission, the extract repeats a garbled "Before placing additional material are permitted" before the clean-and-wet sentence	30 minutes and square-joint permission remain Verified. Do not invent the missing words or adopt commentary 300 mm taper